Learning Dysthe Evolution Across Initial Wave Fields

A literature-grounded numerical, FNO, and PINN research protocol

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Abstract

The proposed goal is to learn the evolution of a bounded family of complex water-wave envelopes and establish how far predictions remain accurate on unseen initial fields. Begin with the approved spatial free-surface Dysthe equation, a qualified Fourier reference, and a short-distance Fourier neural operator (FNO). Compare the same architecture with and without a full equation residual; use an ordinary physics-informed neural network (PINN) on a small set of individual problems to diagnose physics training. A function-conditioned PINN is an optional operator comparison. Success means accurate modulation growth, focusing, recurrence and spectral transfer at a measured cost and within a declared validity range. It does not mean all-distance accuracy or a proof of global smoothness. This report reviews primary literature, audits the current implementation, specifies data and acceptance gates, and supplies small independent operator checks. No FNO/PINN was trained and no new long-distance evolution campaign was run for this review.

1 The question and the proposed contribution

> For a specified family of smooth initial free-surface envelopes, can a learned solution operator predict unseen wave fields and their modulation events over longer propagation distances, with controlled error and useful amortized cost relative to a tuned numerical solver?

This question has three separately measurable parts. First, learn a map across initial fields, rather than fitting a single trajectory. Second, determine the distance over which repeated predictions remain qualified. Third, test whether the method retains the particular water-wave effects for which Dysthe corrections are useful. Reproducing known modulation phenomena is validation; a reproducible improvement in generalization, accuracy per unit cost, or reliable prediction distance would be the methodological contribution. Novel wave physics is not assumed.

The first comparison should be a supervised FNO and the same distance-conditioned FNO with a physics loss, commonly called a physics-informed neural operator (PINO). The FNO and PINO literature supports these method classes, but does not establish their performance on this Dysthe problem [10, 11]. An ordinary PINN trained for each initial field answers a different question and incurs new optimization cost for every field. A branch-conditioned physics-informed operator can answer the multi-input question [12, 13].

Water-wave studies provide closer precedents. Ehlers et al. fit a separate complex-output NLS PINN to synthetic sparse-gauge data, with coefficient identification for model mismatch; Dysthe appears as future work [20]. Ehlers, Stender and Hoffmann use an FNO-based PINO with high-order-spectral free-surface residuals to reconstruct phase-resolved surfaces across sensor inputs

[21]. Eeltink et al. combine viscous MNLS with LSTM corrections learned from breaking-wave experiments; their magnitude and spectral-magnitude targets and wave-type-dependent corrections do not validate complex inviscid rollouts [22]. These precedents motivate distinguishing instance fitting from operator generalization, fixed-equation approximation from missing-physics correction, and reconstruction from autonomous propagation. This is a focused methods review, not a claim of first use or an exhaustive priority search.

Three proposed hypotheses make negative results useful:

1. A short-distance FNO generalizes within a bounded initial-data distribution at a declared error tolerance.
2. Adding a correctly evaluated residual improves held-out rollout accuracy or reduces the amount of reference data required, after accounting for its extra cost.
3. Short rollout training extends the reliable prediction distance without suppressing true modulational instability.

Failure of any hypothesis is a legitimate result. None implies a statement about existence of the underlying PDE.

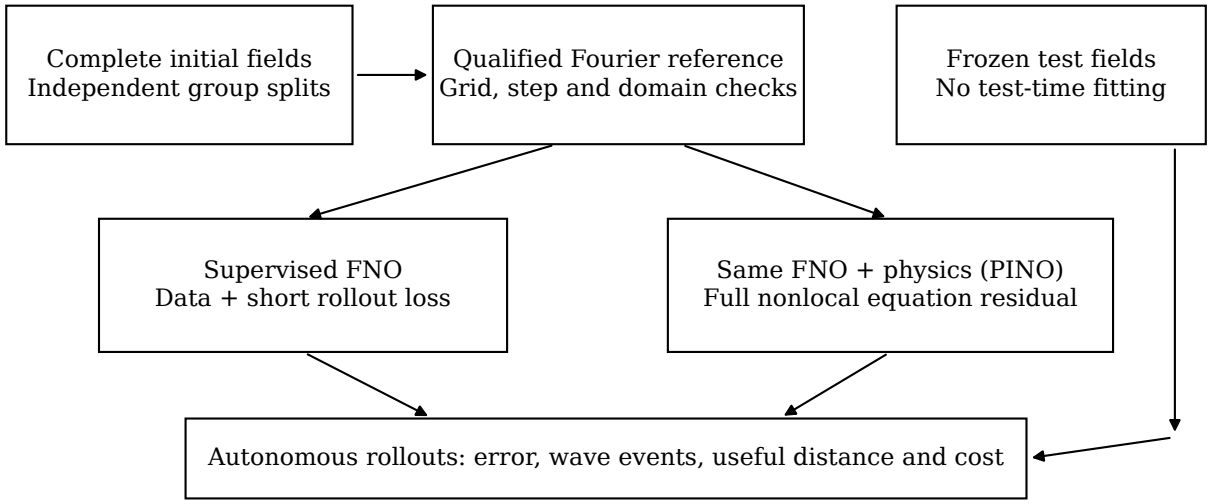


Figure 1: The first controlled comparison. Both models share a representation and data population; only the physics term differs. The ordinary per-input PINN is a separate, smaller diagnostic experiment. This is an experimental design, not a plot of completed results.

2 Fix the equation before comparing methods

On the periodic profile interval $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$, the approved `water-wave-dysthe-spatial-v1` initial-value problem is

$$\begin{aligned}\partial_\xi u = F_\epsilon(u) = &-iu_{\tau\tau} - i|u|^2u \\ &- \epsilon\left(8|u|^2u_\tau + 2u^2\overline{u_\tau} + 2iu|D_\tau|(|u|^2)\right), \qquad u(0,\tau) = u_0(\tau). \end{aligned} \tag{2.1}$$

Here ξ is propagation distance and τ is retarded profile time. This is the free-surface equation with coefficients $(a, h, c, e, f) = (1, 1, 8, 2, 2)$ in Fedele–Dutykh's displayed equation (1.3) [3]. In particular the conjugate derivative term is present. A velocity-potential envelope has different coefficients.

For $k_m = 2\pi m/L$, use

$$\widehat{f}_m = \frac{1}{L}\int_{\mathbb{T}_L} f(\tau)e^{-ik_m\tau}\,\mathrm{d}\tau, \qquad \widehat{|D_\tau| f}_m = |k_m|\widehat{f}_m, \qquad \widehat{|D_\tau| f}_0 = 0.$$

Thus the Hilbert-transform multiplier is $-i\,\mathrm{sign}(k)$ and $\mathcal{H}\partial_\tau = |D_\tau|$. The linear propagator has multiplier $e^{ik_m^2 s}$, not $e^{-ik_m^2 s}$. The stated carrier normalization is $\tau = \epsilon(2x - t)$, $\xi = \epsilon^2 x$, and $B = \epsilon u$. The $\epsilon = 0$ equation is a mathematical NLS control; that change of physical coordinates cannot be inverted at zero.

When a sufficiently regular solution exists on $[0, s]$, write its map as $\mathcal{S}_s^\epsilon : u_0 \mapsto u(s, \cdot)$. This notation presupposes a well-defined solution on that interval. An operator-approximation theorem does not construct that solution or prove its global existence.

2.1 How the two supplied papers enter this project

Dysthe's 1979 paper derives a temporal potential-envelope correction and studies sideband instability [1]. It motivates the mean-flow term and a modulation-growth test. Its variables, dimensionality and coefficients cannot be pasted into (2.1). The supplied Schober–Islas preprint develops a weakly viscous hierarchy and a separate finite-adjustment-time mean-flow closure [2]. It motivates a future, separately specified dissipative study. It neither supplies an operator-learning algorithm nor proves continued smoothness for the present spatial equation.

The earlier companion review, *Dysthe Evolution: Numerical Evidence and Existence* (23 October 2026), records two apparent linear-dispersion inconsistencies in the supplied version 1 and distinguishes its real relaxation closure from a complex damping coefficient. Those issues must be resolved before implementing that extension. The present plan keeps the inviscid spatial model fixed; it does not silently introduce viscosity, transverse coordinates, or a different envelope variable.

2.2 What currently exists

The five active repositories agree on the model and case schema. The local core supplies complex128 Fourier differentiation, projected nonlinear products, integrating-factor RK4, and diagnostics. It keeps $|m| < N/4$, which provides room for its cubic products. The learning registry still marks all methods unimplemented: there is no active water-wave FNO/PINN training pipeline.

Stored short runs at $\epsilon = .025$, $L = 160$, $N = 4096$, $h = .00125$, $\Xi = 4$ include profile, perturbed-profile and Gaussian qualification records. Their largest stored raw relative L^2 grid discrepancy is about 4.2×10^{-15}, step discrepancy 4.1×10^{-9}, and domain discrepancy 2.2×10^{-5}. These are historical two-level comparisons for those cases, not fresh trajectories, error bounds, or evidence for arbitrary inputs. The Gaussian and solitary archive hashes were freshly checked against their reports; the core source hash agrees. A historical Gaussian runner hash differs from today's runner after later formula changes.

A research dataset needs a new bounded generator using the core. The demonstration runner fixes $\epsilon = .025$, $\Xi \le 4$, and at most 101 output stations; part of its initializer also hardcodes that epsilon in provenance. Do not expand the meaning of those records to cover a new parameter campaign. Before packaging a reproducible run, align the learning repository's pinned core dependency with the actual reviewed reference implementation.

3 Multiple inputs: a declared population, not arbitrary waves

Begin with $L = 160$ and $\epsilon = .025$. A training example contains the entire complex initial profile, the actual local propagation increment, and its reference output. Future reference fields are training targets only. The same trained operator must accept a new initial profile without optimization. If adaptation is allowed, label and time it separately.

3.1 Three complementary input families

The following ranges are proposed starting values for a bounded pilot, not ranges established by a physical-validity theorem. Validate a small batch before freezing them. Record all rejected draws and their reason; an unreported rejection rule changes the population being learned.

Localized packets: focusing and asymmetry. Use a periodized version of

$$u_0(\tau) = a \exp\left[-\frac{1-ic}{2\sigma^2}(\tau-\tau_0)^2 + i\phi\right], \quad a \in [.5, 1], \quad \sigma \in [6, 12], \quad c \in [-.5, .5].$$

Vary position and phase as well. Periodize the complex function by summing translated copies, with a documented derivative-level truncation tolerance. A cropped Gaussian is not exactly periodic. This family tests amplitude, width, chirp, focusing distance and asymmetric deformation. Hold out entire parameter draws, not propagation stations. Sech-shaped packets and two separated packets can later be withheld *families*, not silently mixed into an interpolation score.

Modulated wavetrains: instability and recurrence. Use

$$u_0(\tau) = ae^{i\phi}\left[1 + b_+ e^{ik_m\tau} + b_- e^{-ik_m\tau}\right].$$

Choose $a \in [.5, 1]$, integer modes spanning stable, near-boundary and unstable portions of the matrix in Appendix A, and independent sideband phases. Use $|b_\pm|$ between .01 and .05 for a nonlinear pilot; use much smaller paired disturbances for the linearization test. These waves intentionally fill the periodic domain. They do not need a localized-packet edge criterion.

Smooth random profiles: functional variation. Sample complex Fourier coefficients with a prescribed decaying spectrum, for example a finite series with $|m| \le 16$ and weights $\exp[-(m/m_c)^2]$, $m_c \in [4, 10]$, followed by a declared RMS-amplitude choice. Coefficients at m and $-m$ are independent. Bound both peak amplitude and derivative-weighted bandwidth by rejection or a stated sampling rule. This is a finite, smooth input distribution with variable phase, not a claim to represent all sea states. Record its seed and coefficient vector. An alternative spectral law is a separate distribution-shift test.

Information that cannot be discarded. Intensity $|u_0|^2$ alone does not specify the initial-value problem: opposite chirps can have the same intensity and different evolution. Sparse or noisy measurements require an observation model, a reconstruction method and uncertainty evaluation. They are a later inverse problem; they must not be presented as another complete-field input channel with missing phase filled in from the reference.

3.2 Splits, grids and provenance

Store reference fields as complex128 arrays $[N_\xi, N_\tau]$ with $\tau_j = -L/2 + jL/N$, excluding the duplicated endpoint. Store explicit propagation coordinates. The existing runner's rounded output indices can create unequal increments; validate the actual spacing or condition the operator on each actual increment.

Assign one immutable *split group* to each base initial draw. All its windows, refinements, translated or phase-rotated copies, paired disturbances, and coefficient variants remain in the same split. Test for duplicate field hashes as well as mislabeled groups. Generate parameter ranges and seed lists before fitting. Fit global normalization on training fields only. Use validation for model selection; open the final test set once the protocol and checkpoint are frozen. Snapshots from one field are correlated samples, not independent cases.

The current metadata validator is insufficient by itself. The new loader must check actual shapes, dtypes, finite values, coordinates, hashes, model ID, epsilon, grid origin, and qualification status. Preserve case schema 2 and add a separate dataset manifest for group and split records.

Downsample through Fourier restriction, not viewer decimation. Measure projection and reconstruction errors in L^2, H^1, H^2, H^3 over the qualified reference interval. A 512- or 1024-point learning grid is only a candidate; require its representation error to consume at most one tenth of the intended surrogate error budget. Choose the number of learned Fourier modes through a small validation sweep, such as $32, 64, 128$, after inspecting the physical spectra. These numbers do not specify the FNO output bandwidth.

3.3 A campaign that can stop early

Use equal numbers from the three families initially.

Stage	Groups	Train/val/test	Purpose
Execution smoke	12	6/3/3	Load, train, resume and evaluate
Bounded pilot	48	36/6/6	Select viable method and tolerances
Main comparison, if justified	384	288/48/48	Estimate held-out performance

These are proposed budgets, not power calculations or guaranteed sufficient sample sizes. Six pilot test groups cannot substantiate broad generalization. Use learning curves on nested training subsets to decide whether more groups are useful. Keep the test population fixed while increasing training data.

At $N = 4096$ and 81 saved stations on $[0, 4]$, one complex128 trajectory requires about 5.06 MiB before compression. Thus 48 and 384 base trajectories require about 243 MiB and 1.90 GiB, respectively, excluding refinements and checkpoints. A saved interval .05 is a proposed *learning* increment; the reference integration step .00125 is a separate quantity. Extra stations are needed for continuous-distance training and near fast events.

Qualify representative extremes first, then screen every generated trajectory; refine any case whose derivative tails or defects approach the gate. Use full three-level grid and step comparisons on the sentinel panel and difficult cases, and at least an independent refined comparison for accepted evaluation cases. Do not assume the three old demonstrations qualify a whole new distribution.

For the main campaign's first long-distance evaluation, preselect 12 held-out groups (four per family) by input parameters alone. Extend $4 \to 8 \to 16$ and only then, if useful and qualified, to 32. At 641 stations a 12-case $N = 4096$ panel needs about 481 MiB of base fields. The same

panel supports paired comparisons across methods; do not repeatedly select its easiest members. Benchmark actual generation and training costs before launching the main campaign.

Vary epsilon only after the fixed-parameter question works. A proposed second stage uses $.015, .025, .04$ in training and disjoint groups at $.02, .03$ for interpolation. Keep $.05$ a separately labeled extrapolation probe, subject to new qualification and physical screens. Treat $\epsilon = 0$ as an analytic/numerical control. Changing L is a different domain, and changing transverse dimensionality or dissipation is a different model.

4 Implementations that answer the comparison

The first comparison should change the training objective while keeping the representation fixed. Use $\epsilon = .025$ and $L = 160$, and let F_ϵ denote the right-hand side of (2.1). An ordinary PINN approximates one solution after its initial condition has been fixed. An operator accepts an initial *function* and predicts its evolution. Training a coordinate network on several different initial fields without an input identifying the field does not define that operator.

The original FNO learns through Fourier multipliers, pointwise channel maps and nonlinearities; its demonstrations include Burgers, Darcy and Navier–Stokes [10, §§3–5]. PINO adds physics objectives and distinguishes operator training from later optimization for one input [11, §3]. The construction below is our proposed adaptation to spatial water-wave Dysthe. Neither paper establishes its performance for this equation.

4.1 One distance-conditioned Fourier backbone

Let $s \in [0, \Delta]$ be distance within a short propagation slab and $k_m = 2\pi m/L$. Define the known linear propagator by

$$\widehat{S_s u}_m = e^{ik_m^2 s}\widehat{u}_m,$$

which solves $v_s = -iv_{\tau\tau}$. Use the common ansatz

$$G_\theta(u, \epsilon, s) = S_s u + sC_\theta(u, \epsilon, s). \tag{4.1}$$

Here C_θ is a 1D FNO correction, with a declared final Fourier representation. It receives the full initial envelope, the distance s, and, if retained as a channel, the fixed value ϵ. Broadcast scalar conditioning across the profile grid or use a smooth conditioning map. Fourier transforms act only in τ; automatic differentiation acts in s. Propagation distance is not periodic. Use smooth activations and initially double precision for derivative and residual checks.

Concretely, lift the input channels to features z_0 and use blocks

$$z_{\ell+1} = \sigma\Big(W_\ell z_\ell + \mathcal{F}_\tau^{-1}[R_\ell(k)\mathcal{F}_\tau z_\ell] + b_\ell\Big).$$

Here W_ℓ mixes channels pointwise, $R_\ell(k)$ mixes Fourier channels on a chosen finite set of learned modes, and σ is a smooth pointwise nonlinearity. A final channel map and declared Fourier projection produce the two components of C_θ.

This ansatz enforces $G_\theta(u, \epsilon, 0) = u$ for the supplied representation and supplies the correct linear phase. The learned correction can nevertheless be wrong: the linear control must verify its behavior. Setting $\epsilon = 0$ retains the cubic NLS term and is not an isolated linear test. The same lifting, Fourier layers, conditioning, projection and precision must be used in the supervised and physics-informed versions. Architecture changes belong to later ablations, once this comparison has passed the numerical controls.

Represent u by its real and imaginary parts. Do not impose $\widehat{u}_{-m} = \overline{\widehat{u}_m}$ on the complex envelope. Real-valued feature channels have their own Hermitian symmetry, which does not imply that restriction on u. Use an endpoint-excluded grid $\tau_j = \tau_{\min} + jL/N$, $0 \le j < N$, and document whether stored Fourier coefficients use $e^{ik_m\tau}$ or $e^{ik_m(\tau-\tau_{\min})}$. Mixing these conventions introduces a mode-dependent phase error. Keep FFT normalization, mode ordering and the Nyquist treatment consistent with the reference solver.

Fit normalization scales using training data alone. A common scale for both complex channels preserves the meaning of phase rotations. Decode to the physical nondimensional envelope before evaluating F_ϵ; if normalized variables are used directly, transform every coefficient and derivative consistently. Per-input normalization to unit amplitude discards nonlinear strength unless that amplitude is also supplied and restored. Raw absolute position is unnecessary for this translation-invariant periodic equation. Test translation and constant-phase equivariance; a generic two-channel network does not automatically have the latter symmetry.

4.2 A full residual requires a whole profile

For a differentiable reconstructed output $v = G_\theta(u, \epsilon, s)$, define

$$r_\theta(u, s) = \partial_s v - F_\epsilon(v). \tag{4.2}$$

At each sampled s, evaluate a complete periodic slice. Compute $\rho = |v|^2$ and $|D_\tau|\rho$ using the multiplier $|k_m|$, with zero mean mode, and form all derivative nonlinearities. Sparse independent points in (s, τ) cannot determine this nonlocal term. Automatic differentiation through the Fourier evaluation allows training, but does not remove the need to resolve the profile quadrature.

Declare the continuum reconstruction of grid outputs. A convenient choice is an explicit final Fourier projection to $|m| \le K_{\text{out}}$, with the input represented in the same space. Report its initial projection error and refine K_{out}. This final bandwidth is distinct from the number of learned modes in each FNO layer: skip paths and pointwise nonlinearities can generate higher frequencies [11, §3 and Appendix E][15, §§3–4]. For this explicitly truncated output, the cubic right-hand side can reach $3K_{\text{out}}$. A uniform quadrature with $Q > 6K_{\text{out}}$ resolves the full residual and its squared spatial L^2 norm, up to floating-point error. Retain these discarded nonlinear modes in the loss; projecting the residual back to the output band tests only a Galerkin system. This bound does not de-alias arbitrary internal neural nonlinearities. Refine the network evaluation grid separately.

For an endpoint-only FNO, a finite difference or an RK defect in distance measures a discrete update. It is not (4.2) without a specified differentiable reconstruction. Equation (4.1) supplies that distance dependence; Fourier differentiation then refers to the declared profile reconstruction, not an unspecified off-grid network.

4.3 Matched losses and repeated prediction

Write $\|f\|_{2,L}^2 = L^{-1} \int_{\mathbb{T}_L} |f(\tau)|^2 \, \mathrm{d}\tau$. Let $u^{\text{ref}}(s; u)$ be the qualified reference starting from u, and choose fixed positive scales U_*, R_* from training information. For a rollout, set

$$\widetilde{u}_0 = u, \qquad \widetilde{u}_j = G_\theta(\widetilde{u}_{j-1}, \epsilon, \Delta), \quad 1 \le j \le J.$$

A common objective is

$$\mathcal{J}_{\mathrm{data}} = \mathbb{E}_{u,s} \frac{\|G_\theta(u,\epsilon,s) - u^{\mathrm{ref}}(s;u)\|_{2,L}^2}{U_*^2},$$
$$\mathcal{J}_{\mathrm{PDE}} = \mathbb{E}_{u,s} \frac{\|r_\theta(u,s)\|_{2,L}^2}{R_*^2},$$
$$\mathcal{J}_{\mathrm{roll}} = \mathbb{E}_u \frac{1}{J} \sum_{j=1}^{J} \frac{\|\widetilde{u}_j - u^{\mathrm{ref}}(j\Delta;u)\|_{2,L}^2}{U_*^2},$$
$$\mathcal{J} = \mathcal{J}_{\mathrm{data}} + \lambda_r \mathcal{J}_{\mathrm{PDE}} + \lambda_q \mathcal{J}_{\mathrm{roll}}. \tag{4.3}$$

The expectations denote empirical sampling over training inputs and distances; J is a short rollout length. Set $\lambda_r = 0$ for the supervised FNO and $\lambda_r > 0$ for the physics-informed version, with otherwise matched settings. Declare the weights, optimizer, budget and selection rule. Residual distances should include points independent of labeled snapshots. Begin by qualifying single-slab predictions, then introduce the same rollout objective in both models. A physics-only operator is a separate experiment, not the primary comparison.

One-step accuracy does not control inputs produced by the model itself. Brandstetter et al. train on such inputs using a two-step pushforward with gradients stopped through the first step [14, §3.1]. Test this as a named alternative to differentiating through the short rollout in (4.3); do not conflate the two procedures. At evaluation, use predicted endpoints and never reset to the reference. Spectral filtering or bounded weights can reduce numerical sensitivity [15, §4], but global contraction is not an appropriate default for modulationally unstable dynamics. Qualification must retain the reference sideband growth and phase evolution, rather than merely produce bounded or visually smooth trajectories.

4.4 Where a PINN or DeepONet adds information

Use an ordinary PINN first as a per-input diagnostic: on one short slab, represent $v_\theta(s,\tau)$ periodically, enforce its initial value, and minimize the full residual. Its training cost is incurred for each input. A sequential PINN carries its learned endpoint into the next slab; supplying a reference endpoint instead changes the experiment. Published failure studies support testing short intervals and explicit optimization controls [17, §5]. Causal residual weighting prioritizes earlier parts of an interval during training [16, §3]; its weights are optimization diagnostics, not solution-error certificates. These findings motivate the proposed slab strategy but do not guarantee it for Dysthe.

After the shared-backbone comparison passes, an optional function-conditioned spectral DeepONet can test another representation. An encoder receives the initial Fourier coefficients; a smooth distance-conditioned decoder returns complex coefficients $a_m(s;u)$, giving

$$v_\theta(s,\tau;u) = \sum_{|m|\le K_{\mathrm{out}}} a_m(s;u) e^{ik_m(\tau-\tau_{\min})}.$$

Hard initial coefficients and the full residual remain applicable. This is a proposed spectral variant of the branch/trunk idea [12]; physics-informed DeepONet has demonstrated learning across initial functions on other PDEs [13]. Neither universality nor those experiments establish optimization success or generalization to arbitrary water-wave inputs.

Final comparisons use unseen initial realizations, independent profile grids and distances, raw complex errors, derivative norms, and full residuals. Check unweighted residuals even if training uses causal or adaptive weights. Keep phase/translation-aligned errors secondary. Defer test-time

weight adaptation: PINO explicitly permits it [11, §3.2], but it would mix reusable inference with a new optimization solve. If later introduced, report its added cost and accuracy separately. The implementation described here is a proposal; no trained model or long-distance accuracy result is asserted.

5 Which long-distance behavior is worth predicting?

The scientific target is the evolution of a complex wave envelope, including its phase, modulation and transfer between Fourier modes. A useful learned model should preserve these features for unseen initial packets over a declared propagation interval. Agreement with the same Dysthe equation tests the surrogate; agreement with measured water waves tests the physical model. These are separate comparisons.

5.1 Modulation, focusing and recurrence

Dysthe's 1979 calculation identifies induced mean flow as a central correction to modulation-instability predictions [1]. It motivates measuring early sideband growth and the onset of nonlinear exchange. Its temporal potential-envelope formulas must not be substituted directly into the present spatial free-surface equation. Instead, use the linearization of (2.1) and compare small perturbations of a carrier at several modulation frequencies, amplitudes and relative phases. A growing perturbation is evidence of instability of that carrier, not evidence of wave breaking or loss of smoothness.

Localized packets offer a complementary test. Lo and Mei studied asymmetric leaning and splitting, together with unequal sideband growth, using a higher-order envelope model and comparisons with experiments [4]. For Gaussian and sech-like initial packets, measure maximum amplification, the distance and profile location of that maximum, packet asymmetry and spectral broadening. Varying width, chirp and amplitude tests distinct interactions between dispersion and nonlinearity. A small average field error can still hide an incorrect focusing distance or an underestimated peak, so these observables require separate error measures.

Recurrence tests a longer sequence of transfers from carrier to sidebands and back. Armaroli, Brunetti and Kasparian compare full surface-envelope HONLS evolution with a three-mode reduction and distinguish recurrence classes through relative phase [5]. Their reduction is useful as a cheap comparator only while omitted modes remain small; their normalization requires an explicit conversion before quantitative reuse. Measure carrier depletion, sideband imbalance, recurrence distance and the complex field at return. A recurrence defect may minimize over a constant phase and a profile translation, but both fitted shifts and the unaligned error should also be reported. Repeated maxima alone do not establish a breather or accurate recurrence.

Two-packet interactions provide an optional harder test. Fedele and Dutykh numerically observe radiation and changed profiles after spatial-Dysthe collisions, contrasting with their NLS control [3]. Compare the joint evolution with separately propagated packets and quantify outgoing shape, phase, displacement and radiation. The comparison must resolve the small interaction signal and account for overlap; a subtraction artifact is not emitted radiation, and a collision calculation is not a proof of nonintegrability.

5.2 Spectral centroid and spectral peak are different observables

For the Fourier convention $u(\xi,\tau)=\sum_n \widehat{u}_n(\xi)e^{ik_n\tau}$, $k_n=2\pi n/L$, define

$$\overline{k}=\frac{\sum_n k_n|\widehat{u}_n|^2}{\sum_n|\widehat{u}_n|^2},\qquad k_{\mathrm{peak}}\in\arg\max_{k_n}|\widehat{u}_n|^2.$$

The centroid averages all modes; the peak can jump when two modal powers cross. Armaroli et al. show why a temporary mean upshift can coexist with a peak downshift [5]. Neither statistic should substitute for the other.

The following identities are derived directly for our selected baseline. All integrals below are over the periodic profile circle $\mathbb{T}_L$ of length L.

Proposition 5.1 (Centroid constraint for the continuum equation)**.** *Let u be a sufficiently smooth periodic solution of* (2.1) *with real ϵ, and define*

$$A=\int_{\mathbb{T}_L}|u|^2\,\mathrm{d}\tau,\qquad P=\mathrm{Im}\int_{\mathbb{T}_L}\overline{u}\,u_\tau\,\mathrm{d}\tau,\qquad Q=\int_{\mathbb{T}_L}|u|^4\,\mathrm{d}\tau.$$

Then A and $P-\epsilon Q$ are constant. If $A>0$, then

$$\overline{k}(\xi)-\overline{k}(0)=\frac{\epsilon}{A}\big(Q(\xi)-Q(0)\big).\tag{5.1}$$

Proof. Set $\rho=|u|^2$ and $J=\mathrm{Im}\int_{\mathbb{T}_L}\overline{u}^{\,2}u_\tau^2\,\mathrm{d}\tau$. Differentiation and periodic integration by parts give

$$A'=-10\epsilon\int_{\mathbb{T}_L}\rho\rho_\tau\,\mathrm{d}\tau=0,\qquad P'=-4\epsilon J,\qquad Q'=-4J.$$

The mean-flow contribution to P' vanishes because $\int_{\mathbb{T}_L}\rho_\tau|D_\tau|\rho\,\mathrm{d}\tau=0$: $|D_\tau|$ is self-adjoint, commutes with ∂_τ, and preserves real functions. Thus $(P-\epsilon Q)'=0$. Parseval's identity gives $\overline{k}=P/A$, proving (5.1). □

This relation links centroid motion to changes in concentration; it does not prescribe a universal direction of drift. It is a continuum diagnostic, so compare a computed violation with the refined solver's own defect. Projection and time discretization can affect invariant preservation.

Here k is an *envelope* frequency conjugate to τ. With the stated carrier convention $e^{i(x-t)}$ and $\tau=\epsilon(2x-t)$, a mode has physical angular frequency $\omega_0(1+\epsilon k)$ in the positive-carrier band. Changing the Fourier or carrier convention changes this translation. The full reconstructed surface includes bound harmonics; its spectral mean need not equal the first-harmonic envelope statistic.

5.3 What a long calculation can and cannot show

Long propagation means large ξ. Packet width and modulation bandwidth are distinct diagnostics; chirp can broaden the spectrum. The carrier remains a deep-water wave in this model. Moreover, $\xi=\epsilon^2k_0X$ means that the same dimensionless distance corresponds to different physical distances X when steepness varies.

Periodicity also changes the interpretation of long runs. Since A is conserved,

$$\|u(\xi,\cdot)\|_{L^\infty(\mathbb{T}_L)}\geq\sqrt{A/L}.\tag{5.2}$$

Indeed, $A\leq L\|u\|_{L^\infty}^2$. A fixed periodic box therefore cannot demonstrate decay of a nonzero packet to zero. Radiation can return through the boundary; localized-packet conclusions require domain checks and examination of the exterior field.

Track maximum effective steepness and occupied bandwidth as well as numerical resolution. For a conservative pilot screen, one could require $\epsilon \max |u| \le .1$ and $\epsilon k_{95} \le .1$, where k_{95} is the smallest nonnegative cutoff containing 95% of the envelope action in $|k| \le k_{95}$. Record the full tails as well. These proposed screening numbers do not certify asymptotic validity and must be reconsidered for a measured wave population. Trulsen and Dysthe's broader-band extension illustrates that relaxing bandwidth restrictions requires a changed model [6]. No universal amplitude cutoff or indefinitely valid propagation distance follows from a successful envelope computation.

Physical long-distance comparisons need further care. Carter and Govan derive a viscous Dysthe model and compare downshift observables with tank data [7]. A later comparison across four experiments finds that model performance depends on the observable and initial sideband seeding; no model captures every measured evolution [8]. Transverse modulation can also transfer energy and affect downshift in a wide tank [9]. The present inviscid, one-profile equation does not represent dissipation, transverse evolution or breaking.

Finally, accurate finite-distance fields and bounded sampled derivatives support a resolved computational observation. They do not establish global smooth existence. A small training residual does not establish that either: causal PINN training addresses optimization and information propagation [16], while existence and uniform error certification require additional mathematical estimates.

6 Tests with explicit scientific decisions

6.1 Common measurements and acceptance rules

For a reference u and prediction v, measure raw complex errors

$$E_s(\xi) = \frac{\|v(\xi) - u(\xi)\|_{H^s}}{\max(\|u(\xi)\|_{H^s}, \eta_s)}, \qquad \|u\|_{H^s}^2 = L \sum_m (1 + k_m^2)^s |\widehat{u}_m|^2,$$

where η_s is a fixed, declared absolute scale for nearly zero references. Report $s = 0, 1, 2, 3$, maximum amplitude, spectral phase where the signal is above its floor, and spectra. Phase/translation-aligned errors are secondary: raw errors retain mistakes in propagation speed and phase. Record action and modified momentum defects but do not rank methods by conservation alone.

For a continuous reconstruction v, evaluate $r = \partial_\xi v - F_\epsilon(v)$ on independent propagation stations and a refined profile grid, including nonlinear modes outside the output representation. A discrete endpoint defect is labeled separately. Compare residual norms after the same physical normalization and reconstruction; a smaller residual need not imply a smaller field error without a stability estimate.

Proposed first gates are maximum $E_0 \le 1\%$, maximum $E_1 \le 5\%$, and errors at most 5% in the prespecified primary event amplitude and distance. For an event near zero distance, use a declared absolute distance tolerance rather than an ill-conditioned percentage. A regularity interpretation also requires $E_2, E_3 \le 10\%$ and derivative resolution. Require reference/refinement and representation errors to be at least ten times smaller than the corresponding surrogate tolerance. These are engineering targets to freeze after the pilot, not universal physical or rigorous error bounds.

Track the upper fifth of the retained Fourier band in each H^s sum. An initial reference screening target is a weighted-tail fraction below 10^{-6} together with grid agreement; a small tail alone does not pass a case. If a measured Dysthe–NLS difference is smaller than the combined numerical and learning error, the surrogate has not resolved that physical correction. Require those errors below one third of the effect being interpreted, or report the effect as unresolved.

Define the *qualified prediction distance* as the largest tested Ξ for which all applicable gates pass at all checked stations up to Ξ. It is a sampled, tolerance-dependent quantity. Distinguish failure of the surrogate, loss of reference resolution, periodic wrap-around, and departure from physical-model screens. Increase output density around peaks and growth; do not infer a continuous supremum from sparse snapshots.

Report at least three training seeds, individual cases, median and 90th-percentile errors, worst cases and failure counts. Use group-level uncertainty estimates; correlated stations cannot be counted as independent observations. A small pilot needs case-level reporting rather than precise-looking population confidence claims.

6.2 T0. Verify that every method implements the same equation

Goal. Eliminate sign, phase, nonlocal-operator and data-interface errors before interpreting a training experiment.

Experiment and control. Check the linear factor $e^{ik^2 s}$ and the exact nonlinear plane wave

$$u(\xi, \tau) = a \exp\{i[k\tau + (k^2 - a^2 - 6\epsilon a^2 k)\xi]\}.$$

Use a nonconstant two/three-mode field to exercise mean-flow coupling. Check phase/translation equivariance, Fourier-origin reconstruction, normalization inversion, and agreement of the NumPy and differentiable RHS. Check network gradients against directional finite differences. Reject wrong shapes, duplicated endpoints, invalid hashes, split leakage and inconsistent actual output intervals.

Decision and limitation. Algebraic controls should agree to precision-appropriate tolerances before training; double-precision low-mode RHS differences should be near roundoff, with finite-difference gradient errors showing the expected step-size behavior. A failed control blocks the learning comparison. The supplied independent checks are only part of T0: the future differentiable adapter still needs to pass against the core.

6.3 T1. Qualify the numerical reference and representation

Goal. Establish that the target trajectories and learning grid resolve the quantities the research intends to predict.

Experiment and control. Use representative low/high amplitudes, narrow/wide packets, strong chirps and unstable sidebands. Compare $N, 2N, 4N$ at fixed L, and $h, h/2, h/4$ on a sufficiently fine grid. Check expected RK4 step convergence before roundoff dominates. For localized packets compare $L, 2L$ and, if necessary, $4L$ at fixed spacing, reconstructing the same localized field instead of periodically repeating it. Compare common windows, derivative tails and mean flow. Perform the learning-grid projection check separately. Fourier methods have direct spatial Dysthe precedent [3, 4]; the specific gates here are proposed.

Decision and limitation. Accept only cases passing the common reference budget and spectral checks. A larger domain can alter nonlocal mean flow even when edge amplitudes are tiny. An unresolved case triggers refinement or a shorter qualified distance, not a claim of singularity. Full-period wavetrains have a different boundary question from localized packets.

6.4 T2. Learn unseen initial fields at fixed parameters

Goal. Determine whether the FNO learns the declared short-distance operator rather than memorizing a small collection of trajectories.

Experiment and control. Train on the three-family group split at fixed epsilon and length. Evaluate all held-out profiles without retraining. Compare one-step and autonomous rollout errors to a coarser spectral solver whose step and grid are tuned on validation for the same accuracy. Include the exact linear propagator and independently resolved NLS evolution as physical controls, while recognizing that neither solves the full Dysthe problem. Use nested training subsets to produce a learning curve [10].

Decision and limitation. Pass only if the prespecified held-out errors and failure fraction meet the goal. A conservative first main-campaign target is at least 90% of test groups passing the primary gates through $\Xi = 4$, with uncertainty and all failures reported. This establishes bounded-distribution interpolation only. If a cheap spectral solve dominates accuracy and cost, report that outcome before adding model complexity.

6.5 T3. Does the full physics loss actually help?

Goal. Isolate the benefit, if any, of physics training for this equation and input population.

Experiment and control. Compare the same initialized FNO backbone, data, hard initial condition, linear skip, rollout schedule and validation protocol with $\lambda_r = 0$ and $\lambda_r > 0$. Select loss weights using validation only. Report both a matched-compute comparison and the extra cost to reach a matched accuracy; physics updates cost more. Evaluate on new collocation stations and refined full-profile residuals. At a fixed architecture, repeat on two or three data budgets to test data efficiency [11].

Decision and limitation. Physics is useful if it improves the intended held-out metric or needs less reference data at acceptable total cost. A lower training residual alone fails this decision. A physics-only branch is optional after the hybrid works; it is not necessary to launch a large architecture search. If benefits disappear under independent residual refinement, investigate aliasing first.

6.6 T4. What can a PINN contribute?

Goal. Separate single-problem optimization from genuine multi-input physics-informed prediction.

Experiment and control. Choose six qualified validation cases, two per family. For an ordinary Fourier PINN, compare a single interval, sequential distance slabs, and slabs with causal residual weighting at comparable optimization budgets [16, 17]. Carry predicted endpoints forward; do not reset to exact interior snapshots. If this reveals a useful training strategy, test a function-conditioned Fourier/DeepONet model on unseen initial fields with no optimization [13].

Decision and limitation. Count per-case optimization, failed restarts and all slabs. A PINN can be informative without beating the spectral solver. The six-case exercise is a training diagnostic, not a broad statistical comparison. A parameter-conditioned network only generalizes over its declared

packet parameters; an operator claim requires initial-field conditioning and a disjoint field-level test set.

6.7 T5. How far does autonomous prediction remain useful?

Goal. Measure a defensible accuracy horizon and whether rollout training extends it.

Experiment and control. Train on reference intervals no longer than $\Xi = 4$ and repeat learned local maps on their own outputs to $\Xi = 8, 16$, then conditionally 32. Compare one-step training, a short multi-step rollout loss, and optionally stopped-gradient pushforward training [14]. Use validation to select the rollout depth before testing. Plot error and the qualified distance per held-out group. Inspect spectral-error growth, not only norm drift [15].

Decision and limitation. Claim an extension only for the fixed test population and tolerances. Never reset the rollout with a reference state. For a locally Lipschitz exact map the conditional bound $e_{n+1} \leq L_n e_n + \delta_n$ explains why small one-step errors δ_n may accumulate. Here $\delta_n = \|G_\theta(\widetilde{u}_n, \epsilon, \Delta) - \mathcal{S}^\epsilon_\Delta(\widetilde{u}_n)\|$ is the error at a *predicted* input; teacher-forced errors do not supply this bound. This is not a proved uniform Dysthe estimate. Do not enforce artificial contraction merely to keep output bounded.

6.8 T6. Preserve real modulational instability

Goal. Determine whether a learned method reproduces physical sideband amplification instead of damping it away.

Experiment and control. For amplitudes and modulation modes covering stable and unstable regions, compare the full signed perturbation response with $\exp(\xi M(k))$ from Appendix A, a refined Dysthe trajectory, and the NLS limit. Halve the perturbation until the early response converges to the linear prediction while remaining above numerical noise. Check both sideband amplitudes, their phases and growth rates. The observable is motivated by Dysthe's instability calculation [1].

Decision and limitation. Use a provisional 5% gain/growth tolerance over the verified linear interval, with absolute tolerances near zero growth. Require the perturbation signal to exceed the reference error by a declared margin. A filter that keeps a rollout smooth but suppresses the correct growing mode fails. Linear amplification is neither nonlinear blow-up nor wave breaking.

6.9 T7. Predict focusing and recurrence events

Goal. Test the wave behavior that matters beyond average field error.

Experiment and control. For localized packets measure the first focus distance, amplification, peak trajectory and asymmetry against refined Dysthe; compare NLS to identify the size of the higher-order effect [4]. For periodic modulations measure carrier depletion, sideband exchange and recurrence distance [5]. Record raw complex recurrence defect and separately minimize over translation and constant phase. If no event occurs in the qualified reference interval, label it absent/censored rather than forcing a fit.

Decision and limitation. Prespecify event definitions and the 5% primary event target. Similar peak oscillations do not establish recurrence of the complex wave field or a stable breather. Track spectral mean and peak separately and test the exact centroid identity (5.1). Two-packet collisions, with separated single-packet controls and a larger-domain check, are a later family-transfer test; they are not required to establish the first operator result.

6.10 T8. State exactly which generalization is being tested

Goal. Find the boundary of the learned map's competence.

Experiment and control. Change one factor at a time: withheld packet family; new epsilon values within the training interval; epsilon outside it; finer grids on the same domain; a different domain length; longer propagation. Each receives a separate score and qualification. If testing measurement noise later, specify the observation and reconstruction processes and compare to an appropriate estimator, not to a solver supplied with inaccessible noiseless phase.

Decision and limitation. An interpolation result does not imply extrapolation. A finer output grid does not recover missing input frequencies. A domain change alters physical wavenumbers and periodic mean flow. Do not pool these tasks into one favorable average or declare arbitrary-input performance.

6.11 T9. Is there an amortized computational benefit?

Goal. Determine whether the learned solver saves work for the intended repeated-query workload.

Experiment and control. Compare against the fastest numerical configuration that passes the same field, derivative and event gates. Measure hardware, precision, batch size, peak memory, output density, warm-up, inference, data generation, validation, all training seeds and any test-time optimization. Report accuracy–cost curves, not one speed ratio. For a fixed workload a useful accounting identity is

$$M_* = \frac{C_{\text{data}} + C_{\text{train}}}{C_{\text{numerical}} - C_{\text{inference}}},$$

provided the denominator is positive; include other attributable setup/selection costs in the numerator.

Decision and limitation. The estimated break-even query count is meaningful only at matched accuracy and for the specified hardware/batching regime. If the denominator is nonpositive, there is no amortized inference advantage under that comparison. A differentiable surrogate may still be useful, but sensitivity accuracy then becomes a separate test.

7 Implementation order and what this review verifies

First deliverable: a reproducible numerical population. Add a separate research generator, array-aware validation, immutable initial-group manifests and spectral coarsening. Store source/array hashes, the exact model ID, epsilon, coordinates, generation seeds and qualification records. Exercise an interrupted/resumed small generation run. Existing data can supply controls, but cannot substitute for the new distribution or a frozen test split.

Second deliverable: one honest operator comparison. Add the differentiable RHS adapter and a distance-conditioned FNO. Use a small declared architecture, for example four Fourier blocks of width 32 or 64 with smooth activation; choose only a few mode/width settings on validation. Start with the 48-group pilot and the supervised objective. Add the full residual to the same backbone and verify the six ordinary PINN diagnostic cases. Store best and last checkpoints, optimizer state, random-generator state, normalization, training groups and code versions. Check that resumed optimization reproduces the uninterrupted run within the declared determinism tolerance. Never select the best test seed or hide failed runs.

Third deliverable: a qualified behavior comparison. Only after the pilot passes, enlarge the dataset and evaluate the locked long-distance panel. Publish per-case errors, event definitions, unresolved cases, horizon limits and matched-accuracy costs. A useful result may be that physics improves data efficiency but adds inference-independent training cost, that rollout training helps only a subset of inputs, or that the spectral solver remains the better computational choice. The experiment is designed to distinguish these outcomes rather than presume a neural advantage.

Existence remains a separate mathematical task. The relevant computational question is whether the selected initial data continue to exhibit resolved bounded derivatives over the tested distance. An all-distance smoothness claim needs a local theory and continuation criterion for this exact formulation, followed by the necessary uniform bounds, or an appropriate validated a posteriori argument for a specified finite interval. Temporal whole-space Dysthe theory [18] does not automatically apply to this periodic spatial equation. The residual must be combined with the relevant stability, regularity and quadrature estimates [19]; independent training and agreement of two solvers do not supply those estimates. A theorem project should begin with one trajectory and one short interval after the numerical pipeline is qualified.

Checks actually executed for this report. The accompanying `implementation_checks.py` independently evaluates smooth low-mode formulas in NumPy; it does not import or modify the active solver. All checks passed:

Arithmetic check	Largest reported discrepancy
Fourier-origin reconstruction	2.90×10^{-16}
Linear plane-wave phase	1.35×10^{-15}
Linear semigroup composition	2.35×10^{-16}
RHS phase/translation equivariance	6.12×10^{-13}
Real/imaginary versus complex RHS	2.08×10^{-13}
Action directional balance	3.49×10^{-16}
Modified momentum directional balance	7.46×10^{-16}
Centered finite-difference modulation matrix	1.00×10^{-10}

The matrix check spans 21 combinations of $\epsilon = 0, .025, .05$ and integer modes $-3, \ldots, 3$ at $a = .8$. Its discrepancy is consistent with the finite-difference truncation scale. Omitting the grid-origin correction yields a relative field error of about .709 on the deterministic test field. The JSON record contains individual cases, scales, tolerances and software versions; the table combines distinct error types and is not an accuracy ranking.

These checks support implementation of the proposed formulas. They do not complete T0 for a future learning backend or T1 for a new trajectory. No fresh long-distance integrations, trained-model comparisons, physical experiments or existence certificates are reported. Historical run evidence and new arithmetic checks are kept distinct throughout.

A A direct modulation calculation for this convention

This calculation is derived from (2.1); it is not a transcription of a growth formula in another normalization. For $a > 0$ write $u = a(1 + p + iq)e^{-ia^2\xi}$ with real infinitesimal p, q. Keeping only first-order terms gives

$$p_\xi = q_{\tau\tau} - 10\epsilon a^2 p_\tau,$$
$$q_\xi = -p_{\tau\tau} - 2a^2 p - 6\epsilon a^2 q_\tau - 4\epsilon a^2 |D_\tau| p.$$

Consequently each Fourier component satisfies

$$\frac{\mathrm{d}}{\mathrm{d}\xi}\begin{pmatrix}\widehat{p}\\ \widehat{q}\end{pmatrix} = M(k)\begin{pmatrix}\widehat{p}\\ \widehat{q}\end{pmatrix}, \qquad M(k) = \begin{pmatrix}-10i\epsilon a^2 k & -k^2\\ k^2 - 2a^2 - 4\epsilon a^2|k| & -6i\epsilon a^2 k\end{pmatrix}. \tag{A.1}$$

Use $\exp(\xi M(k))$ for the complete small-signal comparison, including phase and both sidebands; fitting a scalar growth slope alone discards information. Its eigenvalues are

$$-8i\epsilon a^2 k \ \pm \sqrt{k^2\left(2a^2 + 4\epsilon a^2|k| - k^2 - 4\epsilon^2 a^4\right)}.$$

The ϵ^2 term arises from diagonalizing the truncated PDE exactly; it is not a newly justified second-order water-wave correction. For an asymptotic comparison at Dysthe order, expand consistently. At $\epsilon = 0$, the familiar NLS unstable band is $0 < |k| < \sqrt{2}\,a$ in this convention. A centered finite-difference check of the nonlinear RHS is supplied with this report. An arithmetic check of the matrix is not a nonlinear stability theorem.

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